CP–7968

Sub. Code
11T

B.Sc. DEGREE EXAMINATION, NOVEMBER 2017

Nautical Science

தாள் 1 – தமிழ்ச் செம்மொழியும் தமிழர்களின் பன்முகத்திறனும்

(2016 onwards)

Time : 3 Hours Maximum : 75 Marks

பகுதி அ (10 × 2 = 20)

எல்லா வினாக்களுக்கும் விடை தருக.

1. உலகச் செம்மொழிகள் எனப்படுபவை எவை?

2. தமிழின் தொன்மைக்கு இரு சான்றுகள் தருக.

3. வண்ண ஆடை என்பதனை விளக்குக.

4. 'போர்வை' – விளக்கம் தருக.

5. வளையல் குறிப்பு வரைக.

6. ஆண்கள் அணியும் அணிகலன்கள் எவை?

7. 'யாழ்' விளக்கம் கூறி வகைகளைத் தருக.

8. ஓவியங்களில் காணப்படும் வனப்பு – விளக்குக.

9. கடற்கலங்களாகச் சங்ககாலத்தில் பயன்பட்டவற்றில் ஏதேனும் நான்கின் பெயர் தருக.

10. நாழிகை கணக்கர் சொற்பொருள் விளக்கம் கூறுக.

பகுதி ஆ $(5 \times 5 = 25)$

எல்லா வினாக்களுக்கும் விடை எழுதுக.

11. (அ) மொழிக்குடும்பங்கள் என்பது பற்றி விளக்குக.

(அல்லது)

(ஆ) தமிழ் மொழியின் சிறப்புக்களைத் தெளிவுப்படுத்துக.

12. (அ) சங்ககால ஆடை வகைகளைப் புலப்படுத்துக.

(அல்லது)

(ஆ) ஆடை அணியும் பண்பாடு பற்றி விவரிக்க.

13. (அ) 'கிண்கிணி' என்பதனை விளக்குக.

(அல்லது)

(ஆ) பெண்கள் அணிந்த பழங்கால அணிகலன்கள் பற்றி குறிப்பு வரைக.

14. (அ) நடனக்கலையின் மாண்புகளை விவரிக்க.

(அல்லது)

(ஆ) சிற்பக் கலையின் சிறப்புகளைத் தருக.

15. (அ) பழந்தமிழரின் அறிவியல் அறிவிற்கு இரு சான்றுகள் தந்து விளக்குக.

(அல்லது)

(ஆ) பண்டைத் தமிழரின் பொருளாதாரம் பற்றி எழுதுக.

பகுதி இ (3 × 10 = 30)

எல்லா வினாக்களுக்கும் விடை தருக.

16. (அ) செம்மொழிக்கான தகுதிகளை வரையறை செய்க.

(அல்லது)

(ஆ) செம்மொழித் தமிழ் நூல்கள் எவை? அவற்றைப் பற்றிய குறிப்பு வரைக.

17. (அ) பெண்கள் மற்றும் குழந்தைகள் அணியும் ஆடைகள் பற்றி புலப்படுத்துக.

(அல்லது)

(ஆ) இசைக்கலையின் வளர்ச்சி பற்றி ஆய்க.

18. (அ) கட்டடக்கலையின் தோற்றம் வளர்ச்சி குறித்து எழுதுக.

(அல்லது)

(ஆ) தமிழர்களின் விருந்தோம்பல் பண்பாட்டை மதிப்பிடுக.

CP–7969

Sub. Code
11H

B.Sc. DEGREE EXAMINATION, NOVEMBER 2017

Nautical Science

Part I – Hindi

HINDI – PAPER I (STORY, NOVEL, GRAMMAR AND TRANSLATION)

(2016 onwards)

Time : 3 Hours Maximum : 75 Marks

Part A $(10 \times 2 = 20)$

सभी प्रश्नों के उत्तर संक्षेप में लिखिए।

1. प्रेमचन्द के किन्हीं चार उपन्यासों का नाम लिखिए।

2. भगवतीचरण वर्मा की प्रमुख रचनाएँ क्या है?

3. 'उसने कहा था' कहानी में किसने किससे क्या कहा?

4. 'पाँच मिनट' कहानी का संदेश क्या है?

5. लिंग किसे कहते हैं? उनके भेदों का नाम मात्र लिखिए।

6. वचन किसे कहते हैं? उनके भेदों का नाम मात्र लिखिए।

7. निर्मला का बचपन कैसे बीता?

8. मन्साराम का स्वभाव कैसा था?

9. भालचन्द्र सिन्हा किस विभाग में काम करती थे? और उनके बेटे का नाम क्या है?

10. रंगीली बाई का संक्षिप्त परिचय दीजिए।

Part B $(5 \times 5 = 25)$

सभी प्रश्नों के उत्तर संक्षेप में लिखिए।

उत्तर 100 शब्दों से अधिक का न हो।

11. (a) आनन्दी का संक्षिप्त परिचय दीजिए।

या

(b) रामू की बहू पर खून कैसे सवार हो गया?

12. (a) लहनासिंह ने बोधासिंह को कैसे बचाया?

या

(b) 'पाँच मिनट' की मुलाकात में बाप – बेटे के बीच में क्या बातें हुई?

13. (a) मन्साराम की मृत्यु कैसे हुई?

या

(b) सुधा का संक्षिप्त परिचय दीजिए।

14. (a) कृष्णा का संक्षिप्त परिचय दीजिए।

या

(b) पंडित मोटेराम का चरित्र – चित्रण कीजिए।

15. (a) 'संज्ञा' किसे कहते हैं? उनके भेदों को उदाहरण सहित समझाइए।

या

(b) पुल्लिंग से स्त्रीलिंग बनाने के नियमों को उदाहरण सहित समझाइए।

Part C

$(3 \times 10 = 30)$

किन्हीं तीन का उत्तर दीजिए। उत्तर विस्तार से हो।

16. (a) कहानी कला के तत्वों के आधार पर 'प्रायश्चित' कहानी का सारांश लिखिए।

या

(b) 'उसने कहा था' कहानी का सारांश लिखिए।

17. (a) सर्वनाम किसे कहते हैं? उनकी भेदों को उदाहरण सहित समझाइए।

या

(b) विशेषण किसे कहते हैं? उनकी भेदों को उदाहरण सहित समझाइए।

18. (a) अंग्रेज़ी में अनुवाद कीजिए।

कौए के संबन्ध में एक रोचक कथा है। कहते है, एक बार वह कोयल का बाना धारण करके चारों ओर लोगों की आँखों में धूल झोंकता किए। कई मास ऐसे ही बीते। जब वसंत आया, तो असली कोयल ने पंचम स्वर में गाना आरंभ कर दिया, पर कौआ चुप रहा। फिर तो उसकी कलई खुल गई और लोग उसे दुतकारने लगे। कहा भी है कि देखने में एक – सा होने पर भी वसंत के आते ही यह साफ़ हो जाता है कि कौआ ही है, कोयल कोयल; यानी बगैर गुण के महज़ ढोंग करने से कुछ नहीं होता।

या

(b) सर्वनाम किसे कहते हैं? उनकी भेदों को उदाहरण सहित समझाइए।

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12

B.Sc. DEGREE EXAMINATION, NOVEMBER 2017

Nautical Science

Part II — ENGLISH — PROSE AND COMMUNICATION SKILLS

(2016 onwards)

Time : 3 Hours Maximum : 75 Marks

Part A (10 × 2 = 20)

Answer **all** questions.

1. What type of education is very important?

2. What are the types of courage that Churchill talks about?

3. Whose experiences does Gardiner narrate?

4. What is the theme of the novel by R.K. Narayan?

5. What does Seton's stories describe.

6. What does the joining of the two rivers symbolize?

7. How many types of tenses are there?

8. Define Adverb with an example.

9. Fill in with Articles :

   He is ———————— architect and ———————— mechanic.

10. ———————— River flows by ———————— side of the city.

Part B (5 × 5 = 25)

Answer **all** questions.

11. (a) Write a note on creative and intellectual Education.

Or

(b) Why does Indira Gandhi disapprove of laying down one's life? What does she ask for, instead?

12. (a) Bring out the humour in the allusion to Stevenson and his Modes in "On Habits".

Or

(b) How does the teacher try to repair the damage done and with what result in "Crime and Punishment"?

13. (a) What is the short coming of new curricula, according to Margaret Atwood?

Or

(b) What does Sarojini Nadiu say about the vision of Religion or Faith?

14. (a) Find out the Sentence Pattern :

(i) I will win the race tomorrow.

(ii) I hate junk food.

(iii) My sister gave a water bottle for me.

Fill in with Modals :

(iv) When ———————— we see you again?

(v) Children ———————— obey their parents.

Or

(b) Fill in the blanks :

(i) I hurt my leg ———————— (fill in with correct adverb).

(ii) ———————— is a sweet smelling flower (fill in with correct noun).

(iii) Napoleon is a ———————— Warrior (fill in with correct adjective).

(iv) Iron and copper ———————— useflul metals (fill in with correct verb).

(v) The river ———————— (flow) under the bridge (use correct tense).

15. (a) Identify the types of Sentences.

(i) He drinks tea every morning.

(ii) When does the coffee house reopen?

(iii) How hot the sun is!

(iv) Get out.

(v) I have found the book.

Or

(b) Fill in with correct Prepositions :

(i) The book is ———————— interesting.

(ii) He sit ———————— on a chair.

Fill in with articles :

(iii) I come the day ———————— yesterday.

(iv) I bought ———————— horse and ———————— ox.

Part C (3 × 10 = 30)

Answer **all** questions.

16. (a) Write an essay on the personality of Indira Gandhi as revealed in her speeches.

Or

(b) What are Gardiner's views on the desirability or otherwise of forming habits.

17. (a) Show how the title "Crime and Punishment" is a satire on the new-fangled notions of teaching that a teacher has to practice.

Or

(b) Fill in with adjectives :

(i) He has ———————— intelligence.

(ii) She is a ———————— girl.

(iii) The ———————— boy was punished

(iv) The ———————— prize was won by Hareesh.

(v) Swimming is a ———————— exercise.

(vi) India expects ———————— man to do his duty.

(vii) He is a ———————— man.

(viii) Hari's mango is ———————— than Ram's.

(ix) Ragu's mango is the ———————— of all.

(x) This is the ———————— temple in Kolkata.

18. (a) In "on Habits", how does A.G. Gardiner use a pencil to talk about the tyranny of habits?

Or

(b) Fill in with Prepositions :

(i) There is a cow ———————— the field.

(ii) The cat jumped ———————— the chair.

(iii) He is fond ———————— tea.

(iv) The soldiers passed ————————.

(v) He is ———————— the room.

(vi) The book lies ———————— the table.

(vii) He jumped ———————— the fence.

(viii) I was searching ———————— this book.

(ix) The path leads ———————— the forest.

(x) He hid ———————— the door.

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13

B.Sc. DEGREE EXAMINATION, NOVEMBER 2017

Nautical Science

NAUTICAL MATHEMATICS – I

(2016 onwards)

Time : 3 Hours Maximum : 75 Marks

Part A (10 × 2 = 20)

Answer **all** questions.

1. Find $\vec{a}+\vec{b}$ if $\vec{a}=3\vec{i}+4\vec{j}-6\vec{k}$ and $\vec{b}=-3\vec{i}+2\vec{i}-2\vec{k}$.

2. Define sample space and events.

3. Write the general equation of circle.

4. Write the standard forms of equation of an ellipse.

5. State the cosine formula in spherical triangle.

6. Define spherical triangle.

7. Differentiate $\frac{\sin x}{x}$.

8. Find $\frac{dy}{dx}$ if $x=a\cos\theta,\ y=b\sin\theta$.

9. Find the rank of the matrix $\begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}$.

10. Define eigen-values of a matrix A.

Part B $(5 \times 5 = 25)$

Answer **all** questions.

11. (a) State and prove Baye's theorem.

Or

(b) An urn contains 4 white balls and 7 black balls. If 6 balls are drawn from the urn, find the probability that among them there will be exactly 2 white and 4 black balls.

12. (a) At which point on the curve $y = x^3 - 12x + 18$ is the tangent parallel to the x-axis?

Or

(b) Evaluate $\int_0^1 \frac{dx}{1+x^2}$, by using Simpson's rule.

13. (a) The sides of a spherical triangle *ABC* are all quadrants and *x, y, z* are the ares joining any point within the triangle to the angular points, prove that $\cos^2 x + \cos^2 y + \cos^2 z = 1$.

Or

(b) In a spherical triangle *ABC*, if $\lfloor B = e$, show that $\sin(A - a) = \sin a \sin A \cos B \cot B$.

14. (a) Differentiate $e^{\sin^{-1} x}$ with respect to $\sin^{-1} x$.

Or

(b) Evaluate $\int \frac{3x+5}{x^2+4x+7} dx$.

15. (a) Show that the vectors $X_1(2, -2, 1)$, $X_2 = (1, 4, -1)$ and $X_3 = (4, 6, -3)$ are linearly independent.

Or

(b) Verify Cayley – Hamilton theorem for the matrix $A = \begin{bmatrix} 6 & -2 & 2 \\ -3 & 3 & -1 \\ 2 & -1 & 3 \end{bmatrix}$.

Part C (3 × 10 = 30)

Answer **all** questions.

16. (a) If the joint probability density function of the random variable X and Y is $f(x,y)=\begin{cases}2-x-y; 0\le x\le 1, 0\le y\le 1\\ 0, \quad elsewhere\end{cases}$, find the correlation coefficient between X and Y.

Or

(b) Find the volume of sphere of radius a.

17. (a) Given $a=36^\circ 28', b=109^\circ 15', c=20^\circ$, solve the triangle.

Or

(b) Evaluate $\int \frac{x}{(x-1)^2(x+2)}dx$.

18. (a) Reduce the quadratic form $3x_1^2-3x_2^2-5x_3^2-2x_1x_2-6x_2x_3-6x_3x_1$ to canonical form.

Or

(b) Find the eigen values and eigen vectors of the matrix $A=\begin{bmatrix}1 & 1 & 3\\ 1 & 5 & 1\\ 3 & 1 & 1\end{bmatrix}$.

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B.Sc. DEGREE EXAMINATION, NOVEMBER 2017

Nautical Science

NAUTICAL PHYSICS AND ELECTRONICS — I

(2016 onwards)

Time : 3 Hours Maximum : 75 Marks

Part A (10 × 2 = 20)

Answer **all** questions.

1. Define centre of mass.

2. What is impulsive force? Write its unit.

3. What are secondary cells?

4. Define RMS.

5. What are isogonic lines?

6. Define co-efficient of viscosity and write its unit.

7. Write a short note on plimsoll mark.

8. What is photoelectric effect?

9. What is meant by crystal diode?

10. Write about beacons.

Part B $(5 \times 5 = 25)$

Answer **all** questions.

11. (a) State and explain Kepler's law.

Or

(b) Explain the working of Weston Differential Pulley.

12. (a) Explain batteries and precautions to be taken.

Or

(b) Write a note on Dip circle.

13. (a) State Stoke's law and derive expression for velocity.

Or

(b) Describe the construction and working of marine hydrometers.

14. (a) Briefly explain photovoltaic cell.

Or

(b) Describe the working of sextant.

15. (a) Explain briefly the working of PN Junction diode.

Or

(b) Discuss about

(i) Radar Altimeters

(ii) Tactical Air Navigation.

Part C (3 × 10 = 30)

Answer **all** questions.

16. (a) Describe an experiment to determine surface tension of a liquid in capillary tube method.

Or

(b) (i) Describe the principle and application of transformer.

(ii) Explain magnetic elements of the earth.

17. (a) State and drive the Bernoulli's theorem.

Or

(b) Discuss the construction and working of Burden pressure gauge.

18. (a) Describe the construction and characteristics of a NPN Transistor in CE-mode.

Or

(b) Write an essay about Radar Communication.

———————

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B.Sc. DEGREE EXAMINATION, NOVEMBER 2017

Nautical Science

NAVAL ARCHITECTURE — I

(2016 onwards)

Time : 3 Hours Maximum : 75 Marks

Part A (10 × 2 = 20)

Answer **all** questions.

1. What is relative density? Mention formula.

2. What is displacement of the ship?

3. Where are double bottom tanks located?

4. What is a bilge strum box?

5. What is flare of a ship?

6. What is dead weight?

7. Mention six degrees of freedom.

8. Mention two advantages of welding over riveting.

9. What is trim?

10. What is the difference between bulk carrier and general cargo vessel?

Part B (5 × 5 = 25)

Answer **all** questions.

11. (a) Explain sheer and camber.

Or

(b) Draw plimsoll line and explain the purpose of load line.

12. (a) Explain FWA and DWA. Mention the formula.

Or

(b) A box shaped vessel 24*5*3m has an even keel draft of 1.2m in DW of RD 1.009. Calculate the draft in DW of RD 1.019.

13. (a) Write short note on peak tanks.

Or

(b) Write any five cargo gears.

14. (a) A ship is floating in FW at a draft of 6.8 m. If the maximum FW draft is 7.0 m, and the SW TPC is 40, find the DWT available.

Or

(b) Explain states of equilibrium.

15. (a) In a ship of 12300 t displacement, KG 10 m, 300 t of cargo was discharged from the lower hold (KG 2 M). Find the final KG.

Or

(b) On a ship of W 6000 t, KM 7.4 m, KG 6.6 m, a double bottom tank of 'i' 1166 m^4 is partly full of HFO of RD 0.95, KM 10.1 m, KG 9.0 m.

Part C (3 × 10 = 30)

Answer **all** questions.

16. (a) On a ship of 15000t displacement KG 7.9 m, a weight of 200t is loaded on the UD (KG 12 m) using the ship's stulken derrick whose head is 30 m above the keel. Find the KG of the ship (i) while loading and (ii) after loading.

Or

(b) Explain :

(i) Centre of gravity

(ii) Righting moment

(iii) List

(iv) Heel

(v) TPC.

17. (a) Explain electric arc welding.

Or

(b) Explain types of framing.

18. (a) Explain :

(i) LOA

(ii) Freeboard

(iii) Breadth

(iv) Draft

(v) Archimedes principle.

Or

(b) A ship of 8000t displacement has KB 3.5 m, KM 6.5 m, and KG 6 m. Find the moment of statical stability at 20° heel, assuming that the deck edge remains above water (ie, ship is still wall – sided at that angle of heel).

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16

B.Sc. DEGREE EXAMINATION, NOVEMBER 2017

Nautical Science

SHIP OPERATION TECHNOLOGY – I

(2016 onwards)

Time : 3 Hours Maximum : 75 Marks

Part A (10 × 2 = 20)

Answer **all** questions.

1. Draw and name the parts of the ship.
   (a) Bulbous bow
   (b) LOA
   (c) LBP
   (d) Rudder.
2. Write down five PPE.
3. Write any five types of merchant vessel.
4. Types of life boat.
5. Types of paint brush.
6. Marking of life budy.
7. How many pyrotechnics are there in life boat?
8. What is the use of immerson suit?
9. What are the types of fire nozzle?
10. Write down the types of ropes.

Part B (5 × 5 = 25)

Answer **all** questions.

11. (a) How will you prepare the surface for painting?

Or

(b) Mark the boxing of compass from North to South.

12. (a) Write down the equipments on life boat.

Or

(b) Write down the look out duties on bridge.

13. (a) Write down the markings of life boat.

Or

(b) What are the LSA requirement on cargo ship.

14. (a) Draw international shore Loupling and Lable IT.

Or

(b) What are the classes of fire and ITS suitable extinguisher?

15. (a) What are the methods of extinguishing fire?

Or

(b) Write down the care and maintenance of wire rope.

Part C (3 × 10 = 30)

Answer **all** questions.

16. (a) Write the definition of NUC, RAM, ICBD, making way, under way.

Or

(b) Write down the handing over duties.

17. (a) What are the safety precautions while lowering life boat?

Or

(b) Explain the following life saving appliance,

(i) Epirb

(ii) Sart

(iii) Pyrotechnis

(iv) Immerson solit

(v) Mob marker.

18. (a) Draw the line diagram of Inert gas system.

Or

(b) What are the maintenance of LSA and FFA equipments?

CP–7975

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21T

B.Sc. DEGREE EXAMINATION, NOVEMBER 2017.

Nautical Science

Paper I — இலக்கணமும் படைப்பிலக்கியமும்

(2016 onwards)

Time : 3 Hours Maximum : 75 Marks

பகுதி அ (10 × 2 = 20)

எல்லா வினாக்களுக்கும் விடை தருக.

1. முதல் எழுத்துக்கள் எனப்படுபவை எவை?

2. மெய்ம்மயக்கம் என்றால் என்ன?

3. 'பாரதியார்' - குறிப்பு வரைக.

4. 'மீராவின்' நூல்கள் இரண்டின் பெயர் தருக.

5. 'புதுமைப்பித்தன்' - சிறு குறிப்பு வரைக.

6. 'அய்க்கண்' சிறுதைக் கருவை நும்பாடப்பகுதி வழி விளக்குக.

7. 'இணையம்' விளக்கு.

8. (Watsup) கட்செவி அஞ்சல் விளக்கம் தருக.

9. கவிதையின் இலக்கணம் தருக.

10. மொழியின் பயன்பாடு யாது?

பகுதி ஆ $(5 \times 5 = 25)$

எல்லா வினாக்களுக்கும் விடை தருக.

11. (அ) மொழி முதல் எழுத்துக்களைச் சான்றுடன் விவரிக்க.

(அல்லது)

(ஆ) வல்லினம் மிகும் இடங்களைச் சான்றுடன் தருக.

12. (அ) 'புத்தகச்சாலை' எனும் தலைப்பில் பாரதிதாசன் கூறுவன யாவை?

(அல்லது)

(ஆ) விதைச்சோளம் எனும் தலைப்பில் கவிஞர் கூறுவன யாவை?

13. (அ) சிறுகதை இலக்கணம் தருக.

(அல்லது)

(ஆ) 'அகிலன்' பற்றிய குறிப்பு வரைக.

14. (அ) வலைப்பூ பற்றி குறிப்பு வரைக.

(அல்லது)

(ஆ) இணைய இதழ்கள் - குறிப்பு வரைக.

15. (அ) சிறுகதையின் உத்திகளை விவரிக்க.

(அல்லது)

(ஆ) இணைய வேலைவாய்ப்பு மையங்கள் - விளக்குக.

பகுதி இ $(3 \times 10 = 30)$

எல்லா வினாக்களுக்கும் விடை தருக.

16. (அ) சார்பெழுத்துக்கள் என்பதனைச் சான்றுடன் விளக்குக.

(அல்லது)

(ஆ) கவிதையின் தோற்றம் வளர்ச்சி பற்றி எழுதுக.

17. (அ) கு.பா.ரா-வின் சிறுகதைத் தலைப்புப் பொருத்தத்தை மதிப்பீடு செய்க.

(அல்லது)

(ஆ) மின்னஞ்சலும், மின் நூலகமும் பற்றி எழுதுக.

18. (அ) இணையத்தில் தமிழ்ப் பண்பாடும் என்பதனை தெளிவுப்படுத்துக.

(அல்லது)

(ஆ) 'அறிவியல் ஆற்றல்' எனும் தலைப்பில் 20 வரிகளுக்கு மிகாமல் கவிதை எழுதுக.

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21H

B.Sc. (Nautical Science) DEGREE EXAMINATION, NOVEMBER 2017

Part I - Hindi – PROSE, GRAMMAR AND TRANSLATION II

(2016 onwards)

Time : 3 Hours Maximum : 75 Marks

Part A (10 × 2 = 20)

सभी प्रश्नों के उत्तर संक्षेप में लिखिए।

1. वैज्ञानिक युग में भारत का दायित्व क्या है?

2. चूड़िहारिन की पेशे में क्या विशेषताएँ होनी चाहिए।

3. घृणा और भय में क्या सम्बन्ध है?

4. युवति कवि की ओर क्यों आकर्षित हुई।

5. लेखक बाढ़ में क्यो और कैसे फँस गया?

6. भारत के सम्बन्ध में गाँधी की कल्पना क्या थी?

7. वह कौन सी वस्तु है जो विभिन्न में एकता बनाये हुए हैं?

8. किस व्यक्ति, वस्तु तथा भाव से घृणा करनी चाहिए?

9. लेखक ने रजिया को अन्तिम बार किस रूप में देखा?

10. मक्रील के वातावरण पर प्रकाश डालिए।

Part B $(5 \times 5 = 25)$

संदर्भ सहित व्याख्या कीजिए।

11. (a) हमारी सामूहिक चेतना ऐसे नैतिक आधार पर ठहरी हुई है, जो पहाड़ों से भी मज़बूत, समुद्रों से भी गहरी और आकाश से भी अधिक व्यापक है।

Or

(b) जिस वस्तु का जीवन में इतना मूल्य हैं, उसे शिथिल होने देना अपने पाँव में कुल्हाडी मारना है।

12. (a) उसका चेहरा अचानक बिजली के बल्ब की तरह चमक उठा और चमक उठीं वे नीली आँखें, जो कोटरों में धँस गयी थीं।

Or

(b) समीप खड़ी एक युवती को देख, भद्रता से टोपी छूते हुए उसने पूछा, "आप भी इसी होटल में ठहरी हैं!"

13. (a) 'रमता राम' इसलिए कहती हैं कि जो रमता नहीं, वह राम नहीं। टिकना मौत है।

Or

(b) आधी – भूखी जनता कोई धर्म नहीं रख सकी, न कोई कला, न कोई संगठन।

14. (a) 'ने' विधी का प्रयोग समझाइए।

Or

(b) अकर्मक क्रिया किसे कहते हैं? उदहरण सहित समझाइए।

15. (a) सकर्मक क्रिया किसे कहते हैं? उदहरण सहित समझाइए।

Or

(b) प्ररणार्थक क्रिया किसे कहते हैं?

Part C $(3 \times 10 = 30)$

किन्हीं तीन प्रश्नों का उत्तर दीजिए। उत्तर विस्तार से हो।

16. (a) कहानी कला की दृष्टि से 'मक्रील' की समीक्षा कीजिए।

Or

(b) 'महात्मा गाँधी' निबंध का संक्षिप्त सार अपने शब्दों में लिखिए।

17. (a) 'कारक' किसे कहते हैं? उनके भेदों को उदाहरण सहित समझाइए।

Or

(b) काल किसे कहते हैं? उनके भेदों को उदाहरण सहित समझाइए।

18. (a) हिन्दी में अनुवाद कीजिए।

Shops advertise in the newspapers the things they have to sell. From the advertisements we learn where to buy such things as food, clothing and furniture. We also find out where to go to hear good music or to see a picture. People advertise in newspapers for help or for work. They advertise about houses to let or for sale. Sometimes in the newspapers there are pictures of interesting events that have happened.

Or

(b) 'बहता पानी निर्मला' पाठ का सारांश लिखिए।

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B.Sc. DEGREE EXAMINATION, NOVEMBER 2017

Nautical Science

Part II – ENGLISH – PROSE AND COMMUNICATION SKILLS

(2016 onwards)

Time : 3 Hours Maximum : 75 Marks

Part A (10 × 2 = 20)

Answer **all** questions.

1. Who were the foreigners who invaded and looted India?
2. Name some of the duties that the artist, as a citizen, has to the community.
3. What are the two ways in which science benefits man?
4. Why does Nehru regard Gandhiji as the sun?
5. How was Lynd converted to nationalism?
6. What is discipline of reality? How does a child learn realities?
7. Who are the two important characters in O' Henry's "After Twenty Years"?
8. What did the conjurer do to the Quick Man's watch?
9. What, according to the astrologer, are the common problems of most people?
10. What are the names of the brothers in "The Four Brothers"

Part B $(5 \times 5 = 25)$

Answer **all** questions.

11. (a) What does Kalam tell about Indian's attitude to burning social issues?

Or

(b) Sum up Haldone's remarks on the Negroes.

12. (a) Write a paragraph on the opening of Nehru's speech.

Or

(b) What are Robert Lynd's experiences as an arguer in school and college?

13. (a) Bring out the humour in Stephen Leacock's "The Conjurer's Revenge".

Or

(b) How does R. K. Narayan sketch the character of the astrologer in his "An Astrologer's Day"?

14. (a) What is the difference between a main clause and a subordinate clause? Give two examples having both main clause and subordinate clause.

Or

(b) How do phrase help in making of simple sentence.

15. (a) Write any five words used in direct speech that gets changed in indirect speech for example "now" changes itself into "then"

Or

(b) What are the three degrees of comparison? Give an example for each.

Part C (3 × 10 = 30)

Answer **all** questions.

16. (a) How does Kalam's "My Vision of India" is a visionary agenda full of hopes?

Or

(b) How does Nehru react to Gandhi's death and how does he finally reconcile himself to it?

17. (a) What role does Quick Man play in the story? If the Quick Man was not there, how would the story end?

Or

(b) What is the irony of the story "An Astrologer's Day"?

18. (a) Fill in with the correct adverbial clause

(i) Don't talk ———————— (when/while) she is singing.

(ii) He came ———————— (since/after) night had fallen.

(iii) Do it ———— (since / before) you forget.

(iv) I am glad ———————— (because / that) you like it.

(v) ———————— (if/unless) you work hard, you will fail.

(vi) You must go ———————— (whether / if) you hear from him or not.

(vii) ———————— (wherever / whatever) happens keep calm.

(viii) He is such a good man ————— (that / so) all respect him.

(ix) ———————— (however/whichever) road we take we shall be too late.

(x) He is older ———————— (because / than) he looks.

Or

(b) Change in to Direct Speech

(i) He asked his father when the next letter would come

(ii) I asked my servant if tea was ready

(iii) The child exclaimed with joy that his mother had come.

(iv) The teacher ordered Raju to leave the room at once.

(v) He told me that he had killed a deer the previous day.

———————

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B.Sc. DEGREE EXAMINATION, NOVEMBER 2017.

Nautical Science

NAUTICAL MATHEMATICS – II

(2016 onwards)

Time : 3 Hours Maximum : 75 Marks

Part A $(10 \times 2 = 20)$

Answer **all** questions.

1. Find the modulus of $\frac{2+i}{(1+i)^2}$.

2. Find the real and imaginary parts of $\sin(x+iy)$.

3. Define numerical integration.

4. Write Simpson's $\frac{1}{3}$ rd rule formula.

5. Find the work done in moving a particle in the force field $\vec{f} = 3x^2\vec{i} + (2xz - y)\vec{j} + z\vec{k}$ along the straight line form $(0,0,0)$ to $(2,1,3)$.

6. Show that the vector $2xy\,\vec{i} + (x^2 + 2yz)\,\vec{j} + (y^2 + 1)\,\vec{k}$ is irrotational.

7. Form the differential equation by eliminating a and b from $xy = ae^x + be^{-x}$.

8. Solve $\frac{dy}{dx} = \frac{2x}{x^2 + y^2 - 2y}$.

9. Solve $(D^3 - 3D^2 + 4)\, y = 0$.

10. Define Whirling speed of a shaft.

Part B $(5 \times 5 = 25)$

Answer **all** questions.

11. (a) Simplify $\frac{(\cos 3\theta + i \sin 3\theta)^4 (\cos 4\theta - i \sin 4\theta)^5}{(\cos 4\theta + i \sin 4\theta)^3 (\cos 5\theta + i \sin 5\theta)^{-4}}$.

Or

(b) Sum the series

$$\sin^2 \theta - \frac{1}{2} \sin 2\theta \sin^2 \theta + \frac{1}{3} \sin 3\theta \sin^3 \theta - \frac{1}{4} \sin 4\theta \sin^4 \theta + \ldots \infty .$$

12. (a) Calculate $\frac{dy}{dx}$ and $\frac{d^2y}{dx^2}$ at $x = 1.35$ from the following

x	1.1	1.2	1.3	1.4	1.5	1.6
y	–1.626	0.156	2.453	5.392	9.125	13.831

Or

(b) Evaluate $\int_0^1 \frac{dx}{1 + x^2}$, using Trapezoidal rule with $h = 0.2$. Hence determine the value of π.

13. (a) If ϕ is a solution of the Laplace equation, prove that $\nabla\phi$ is both solenoidal and irrotational.

Or

(b) Evaluate $\int_C \phi\, d\vec{r}$, where C is the curve $x = t, y = t^2$, $z = (1-t)$ and $\phi = x^2 y(1+z)$ from $t = 0$ to $t = 1$.

14. (a) Solve : $\tan y \sec^2 x\, dx + \tan x \sec^2 y\, dy = 0$.

Or

(b) Solve : $ydx - xdy - 3x^2 y^2 e^{x^3} dx = 0$.

15. (a) Solve : $(D^2 + 5D + 6)\, y = e^x$.

Or

(b) Solve : $(D^2 + D + 1), y = \sin 2x$.

Part C (3 × 10 = 30)

Answer **all** questions.

16. (a) State and prove De Moivre's theorem.

Or

(b) Evaluate $\int_0^1 e^{-x^2} dx$ by dividing the range of integration into 4 equal parts using

(i) Simpson's $\frac{1}{3}$ rule and

(ii) Weddle's rule

17. (a) Evaluate using Green's theorem in the plane for $\int_C (xy + y^2)\, dx + x^2 dy$ where C is the closed curve of the region bounded by $y = x$ and $y = x^2$.

Or

(b) Verify Gauss-divergence theorem for

$\vec{f} = (2x - z)\vec{i} + x^2 y\, \vec{j} - x z^2\, \vec{k}$ over the cube bounded by $x = 0, x = 1, y = 0, y = 1$ and $z = 0, z = 1$.

18. (a) Solve $y(xy + 2x^2 y^2)\, dx + x(xy - x^2 y^2)\, dy = 0$.

Or

(b) Solve : $(D^2 + 1)\, y = x^2 e^{2x}$.

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	24

B.Sc. DEGREE EXAMINATION, NOVEMBER 2017

Nautical Science

NAUTICAL PHYSICS AND ELECTRONICS — II

(2016 onwards)

Time : 3 Hours Maximum : 75 Marks

Part A $(10 \times 2 = 20)$

Answer **all** questions.

1. Define Nuclear fission with examples.

2. What is oil splashing?

3. What is meant by impedance?

4. Define Skip zone.

5. Convert $(15)_{10}$ to $(x)_2$.

6. What are logic gates?

7. Define current gain.

8. Define Modulation and write its types.

9. What is meant by ship borne VHF?

10. Define Selectivity.

Part B $(5 \times 5 = 25)$

Answer **all** questions.

11. (a) Explain charging of oil in pipeline flow with its remedies.

Or

(b) What is satellite and deduce the expression for velocity of it in an earth orbit?

12. (a) Define critical frequency and radiation pattern of Hertz Antenna.

Or

(b) (i) Add $(10011)_2$ and $(1111)_2$.

(ii) Subtract $(1001)_2$ from $(1111)_2$.

13. (a) Explain construction and working of AND gate with neat diagram.

Or

(b) Briefly explain the working of R-S flip flop.

14. (a) Explain transistor as amplifier in CE Mode.

Or

(b) Describe the working of transistor as Switche's device.

15. (a) Briefly explain the frequency modulation.

Or

(b) Give the function of transmitter with block diagram.

Part C (3 × 10 = 30)

Answer **all** questions.

16. (a) Describe the construction and working of Nuclear reactor.

Or

(b) Describe the theory of resonance in LCR circuit in series.

17. (a) NAND and NOR gates are universal gates – Explain with diagram.

Or

(b) Describe the construction and working of Bistable multivibrator with neat diagram.

18. (a) Describe in detail about the superhetrodyne receivers.

Or

(b) Explain about the descriptive treatment of transmission of information of the modulated carrier wave.

CP–7980

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25

B.Sc. DEGREE EXAMINATION, NOVEMBER 2017

Nautical Science

SHIP OPERATION TECHNOLOGY — II

(2016 onwards)

Time : 3 Hours Maximum : 75 Marks

Part A (10 × 2 = 20)

Answer **all** questions.

1. Draw the sketch of draft and air draft.

2. Write down the types of stopper and its uses.

3. What is let go?

4. When you will use combination ladder?

5. Write the full form of FPD and where IT is used.

6. Write the full form of SART and EPIRB.

7. What is the anchor day signal and night signal?

8. Write down the types of portable fire extinguisher.

9. What is the term vessel bought up?

10. Draw the fire triangle.

Part B (5 × 5 = 25)

Answer **all** questions.

11. (a) Explain securing anchor for sea.

Or

(b) What is bitter end, foul anchor, anchor aweigh and anchor sighted and clear?

12. (a) LSA requirement of cargoship from forward to AFT.

Or

(b) What is emergency and types of alarms?

13. (a) Write down the precautions while working aloft?

Or

(b) What are the actions to be taken in the event of fire?

14. (a) FFA requirement of cargo ship from forward to AFT.

Or

(b) How to emergency fire pump manually?

15. (a) Effect of shallow water, squat and bank effect.

Or

(b) Write down the action for man over board.

Part C (3 × 10 = 30)

Answer **all** questions.

16. (a) Draw the diagram of Williamson turn and Schanrow turn.

Or

(b) Write down the operation and refilling extinguisher of CO_2, foam and DCP.

17. (a) Write down the action before and after abandon ship.

Or

(b) State safety precaution on rigging combination ladder. When you will combination ladder on port side 1 mtrs above water level?

18. (a) What are the uses of SART, EPIRB, TPA, Emerson suit, rocket parachute?

Or

(b) Write down the types of Davit and method of operation.

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26

B.Sc. DEGREE EXAMINATION, NOVEMBER 2017

Nautical Science

NAVIGATION – I

(2016 onwards)

Time : 3 Hours Maximum : 75 Marks

Part A (10 × 2 = 20)

Answer **all** questions.

1. What is Equator? Explain with diagram.

2. What is d'lat? Explain with diagram.

3. What is Meridianal parts? Where do you find the value of each latitude?

4. Define Mercator sailing.

5. What is deviation? Explain with diagram.

6. Find the compass course from the following true course: 0.75° (T), variation 5°E, Dev 3°W.

7. What is estimated position? Explain.

8. What do you understand by magnetic variation change in annual value?

9. What is the meaning of the term parallax?

10. Define and explain sextant attitude.

Part B (5 × 5 = 25)

Answer **all** questions.

11. (a) Define the following with neat diagram (i) poles (ii) latitude (iii) Meridians (iv) distance (v) nautical miles. (5)

Or

(b) (i) Explain "units of measurement".

(ii) What is the difference between Geographical mile and Nautical mile?

12. (a) (i) Define parallel sailing.

(ii) Two vessel's on the equator were 60 miles apart. Both steered 180°(T) until they reached latitude 30°S. Find the distance between those two vessels at 30°S.

Or

(b) Find the difference in speed at which two places, one in lat 22°S and the other in lat 43°N are carried out by the Earth's rotation. (5)

13. (a) (i) What is observed position? (1)

(ii) On 29th Nov, AM at ship DR 26° 27' N 130°27'W, an astronomical observation was made when the chron show 05h 46m 20s. If the chron error was 00m 10s fast, find correct GMT. (4)

Or

(b) Find M'lat from the following: (5)

	(i)	(ii)	(iii)	(iv)	(v)
Lat A	10°26'N	46°52'S	20°19'N	37°48'S	20°11.0'N
Lat B	30°50'N	30°10'S	40°29'S	53°36'N	23°32.8N

14. (a) Find d'long from the following: (5)

	(i)	(ii)	(iii)	(iv)	(v)
From:	075°10'E	124°11'W	013°06'E	175°30'E	004°11.3'E
To:	069°08'E	136°14'W	005°33'W	170°20'W	002°10.0'W

Or

(b) On 29th Nov 2008, in DR 26°27'N 130°27'W, the sextant attitude of the sun's upper limb east of the meridian was 28°11.0'. If the IE was 2.3' off the are and HE was 10m, find the true attitude. (5)

15. (a) (i) What is Azimuth. (2)

(ii) What is the procedures for checking accuracy of Azimuth Mirror? (3)

Or

(b) Define the following terms (Explain) with diagram. (5)

(i) Reading of Horizontal and vertical sextant angle.

(ii) Dip and refraction.

Part C (3 × 10 = 30)

Answer **all** questions.

16. (a) Find the course and distance between given two positions.

From Lat: 20°10'N Long: 179°40'W

To Lat: 13°40'N Long: 178°10'E

(Plane sailing). (10)

Or

(b) Find the position arrived from the following information by using plane sailing method: (10)

Position A: Lat 10°12'5 Long: 120°11'W

Course: 260°(T)

Distance: 458M

17. (a) Find the course and distance from given two position (use traverse table) (10)

From: 03° 12.0'N 004° 11.3'E

To: 02° 30.4'S 002° 10.0'W.

Or

(b) Find by Mercator's sailing, the course and distance from the given two positions. (10)

From: 06° 00'N 079° 00'W

To: 38° 00'S 179° 40'E.

18. (a) Find the arrived position by using mercator principle from the following starting position, course and distance

Position A: 36° 48'N 085°53 W

Course: 241° (T)

Distance: 1897 M. (10)

Or

(b) At Noon on 14th Sept, a ship in position 40° 12'N 076° 46'W. Set course of 250°(C), Dev. 4°W, Var. 6°W at an engine speed.

At 1600, course attered 287°, Dev. 2°E

Var. 6°W', and speed reduced to 14 kts.

At 2200, altered course to 340°(C), Dev. 5°E,

Var. 5°W and speed 14 kts.

At 0600, altered course to 277°(C)

Dev. 2°W, var. 5°W, speed increased to 15 kts.

This course maintained till 15th Noon.

Find the EP on 15th Noon, the course and distance made good, if current was setting 027°(T) at 2kts throughout. (10)

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B.Sc. DEGREE EXAMINATION, NOVEMBER 2017.

Nautical Science

ENVIRONMENTAL STUDIES

(2016 onwards)

Time : 3 Hours Maximum : 75 Marks

Part A (10 × 2 = 20)

Answer **all** questions.

1. Natural environment.

2. Earth summit.

3. Tidal energy.

4. Timber extraction.

5. Ex-situ conservation.

6. Trophic level.

7. Genetic diversity.

8. Endemic species.

9. Red data book.

10. Water logging.

Part B $(5 \times 5 = 25)$

Answer **all** questions.

11. (a) Enumerate the scope of environmental studies.

Or

(b) Discuss the impact of industrialization in environment.

12. (a) Write a account on water resources.

Or

(b) Discuss the impact of dams on environment.

13. (a) Explain the types of biodiversity.

Or

(b) Describe the food chain and food web.

14. (a) Give an account on thermal pollution.

Or

(b) Discuss how marine environment is affected by pollution?

15. (a) Describe the threats to biodiversity.

Or

(b) Write an account on threats to biodiversity.

Part C (3 × 10 = 30)

Answer **all** questions.

16. (a) Write an account on renewable and non-renewable energy resources.

Or

(b) Elaborately discuss energy flow in an ecosystem.

17. (a) India is a megadiversity nation-discuss..

Or

(b) Write an essay on biogeographicai classification of India.

18. (a) Describe the impact of air pollution and its sources.

Or

(b) Give a detailed account biodiversity conservation methods and its advantages.

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B.Sc. DEGREE EXAMINATION, NOVEMBER 2017

Nautical Science

Part II — ENGLISH AND COMMUNICATION SKILLS — I

(2016 onwards)

Time : 3 Hours Maximum : 75 Marks

Part A (10 × 2 = 20)

Answer **all** questions.

1. What is verbal communication?

2. How do you encounter the barriers of communication?

3. Frame questions with subject, verb and object.

4. Define conditional Clause.

5. Use appropriate form of prepositions :

   (a) There was a road ———————— the two houses.

   (b) They will come here ———————— 7 O'clock in the morning.

6. Define adjectives with example.

7. What are the types of intonation?

8. Define Primary stress with example.

9. What are SQ3R?

10. What is scanning?

Part B $(5 \times 5 = 25)$

Answer **all** questions.

11. (a) What are the kinds of communication? Explain.

Or

(b) What are the barriers of communication?

12. (a) Give a brief account of Prepositional Phrases.

Or

(b) How do you frame questions using 'Wh' questions?

13. (a) Explain the types of adverbs.

Or

(b) Write about zero articles.

14. (a) How sounds are classified?

Or

(b) Write the importance of secondary stress in the sentences.

15. (a) Write the importance of reading.

Or

(b) Differentiate loud reading and silent reading.

Part C $(3 \times 10 = 30)$

Answer **all** questions.

16. (a) What are the barriers for communication? How will you overcome?

Or

(b) How will you transform Interrogative sentences from Active voice into Passive voice? Explain with examples.

17. (a) Write an essay on Infinitive and gerund.

Or

(b) What are the special features of Good Handwriting?

18. (a) What are the types of prepositions? Give examples.

Or

(b) Write briefly the mechanics of Handwriting.

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B.Sc. DEGREE EXAMINATION, NOVEMBER 2017

Nautical Science

CARGO HANDLING AND STOWAGE — I

(2016 onwards)

Time : 3 Hours Maximum : 75 Marks

Part A (10 × 2 = 20)

Answer **all** questions.

1. Define Bulk carrier.

2. Define Grain capacity.

3. Define Breaking strength.

4. What is cargo sweat?

5. What is the need of battening the cargo before loading deck cargo?

6. What is the minimum angle between the derricks with respect to horizontal and vertical when rigging union purchase?

7. Name the parts of a simple derrick.

8. Name the parts of a container.

9. What is dry freight container?

10. What is reefer ship and explain the type of cargo they carry?

Part B (5 × 5 = 25)

Answer **all** questions.

11. (a) Briefly explain the types of cargo a general cargo ship carry.

Or

(b) What is the importance of cargo care to economical operation of the ship?

12. (a) Draw a neat diagram of a container.

Or

(b) What is Bulk carrier and explain the types of cargo they carry?

13. (a) Explain the following with neat diagram :

(i) Bale capacity

(ii) Grain capacity.

Or

(b) Explain the following ;

(i) Factor of safety

(ii) Breaking strength.

14. (a) Explain the factors which affect sweat in the cargo holds.

Or

(b) Explain briefly what is dunnage and its uses on board ships.

15. (a) Describe shortly about segregation and cave of container carrying dangerous goods.

Or

(b) Explain the following with respect to reefer ship.

(i) Use of brine traps

(ii) Purpose of temperature recording.

Part C (3 × 10 = 30)

Answer **all** questions.

16. (a) Enumerate the procedures adopted in ensuring safe handling of derricks.

Or

(b) List the general principles of cargo stowage and handling required to ensure proper stowage of cargo on board a general cargo ship with regards to the following :

(i) Safety of ship and crew

(ii) Safety of cargo.

17. (a) Discuss elaborately "Trades on which sweat is likely to occur and procedures adopted to counter it"?

Or

(b) Enumerate the contents of lashing code and cargo securing manual.

18. (a) What are the operational checks to be done on ship's crane before handing over to stevedors?

Or

(b) List the methods employed in unitisation of general cargoes.

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B.Sc. DEGREE EXAMINATION, NOVEMBER 2017

Nautical Science

MARINE ENGINEERING CONTROL SYSTEM — I

(2016 Onwards)

Time : 3 Hours Maximum : 75 Marks

Part A (10 × 2 = 20)

Answer **all** questions.

1. What is smelting? What is the use of coke and flux in smelting?

2. How are steel classified?

3. What is the Hydrophor system? What are the advantages of Hydrophor system?

4. Explain briefly the storage and supply of fresh water in ships.

5. What is ventilation? Explain natural ventilation and forced ventilation arrangements. What are the Solas requirements for the inlet and outlet of all ventilations?

6. Explain the types of pumps used on board the vessel. State the working principle of pumps.

7. Explain 'A' frame and where it is fitted in the Main Engine. What is the purpose of Tie Rod What are the consequences of running the engine with loose Tie Rods.

8. Define gas law.

9. Define power factor.

10. What is the functions of circuit breaker?

Part B $(5 \times 5 = 25)$

Answer **all** questions.

11. (a) Give the list of all heat treatment process.

Or

(b) What is blast furnace? Explain in detail about the extraction of iron.

12. (a) Sketch and describe Babcock and Wilcock Header Boiler. Explain the (i) single pass type (ii) Double pass type (iii) Three pass type.

Or

(b) Explain the Storage and distribution of Compressed Air. What are the requirements for air compressors on-board ships?

13. (a) Explain with sketch the working of electro hydraulic steering system. Explain the operation of hunting gear mechanism.

Or

(b) Sketch and describe a centrifugal pump used on board and mark the parts. What is the purpose of wear ring and lantern ring? Explain the uses of centrifugal pumps.

14. (a) Draw the valve timing diagram of four stroke diesel engine and explain the working principle.

Or

(b) Explain the faults detected in the Main engine (i) Early injection (ii) Late injection and after burning (iii) Loss of compression.

15. (a) Explain the conditions for parallel operation.

Or

(b) Explain the importance of Circuit breaker and its types.

Part C $(3 \times 10 = 30)$

Answer **all** questions.

16. (a) Explain the working of positive displacement pumps. Why air vessels is fitted in the discharge pipe. Why relief valve is fitted in this type pumps?

Or

(b) Describe the principle objects of cylinder lubrication and the position of the cylinder lubricants for injection oil. Explain the abrasive liner wear and corrosive liner wear.

17. (a) Sketch and describe water tube boiler used on onboard ship. Explain the various water walls used in the boiler.

Or

(b) Write short notes on the following (i) stainless steel (ii) cast iron (iii) alloy steel (iv) wrought iron (v) brass.

18. (a) Explain the block diagram of power generation, Transmission and distribution system on onboard ship.

Or

(b) What is heat treatment? Explain in detail about the various types of heat treatment.

———————

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B.Sc. DEGREE EXAMINATION, NOVEMBER 2017

Nautical Science

VOYAGE PLANNING AND COLLISION PREVENTION — I

(2016 onwards)

Time : 3 Hours Maximum : 75 Marks

Part A (10 × 2 = 20)

Answer **all** questions.

1. A vessel's position should be plotted using bearings of _____.
   (a) Buoys close at hand
   (b) Fixed known objects on shore
   (c) Buoys at a distance
   (d) All of the above

2. To know the sea bed which instrument is used?
   (a) Radar (b) AIS
   (c) VHF (d) Hand Lead Line

3. Which rule defines safe speed?
   (a) Rule 6 (b) Rule 5
   (c) Rule 1 (d) None of the above

4. In finding out compass error which parameter varies from place to place?
   (a) Deviation (b) Variation
   (c) Sea water (d) Mountains

5. Which rule defines responsibility?
   (a) Rule 5 (b) Rule 6
   (c) Rule 2 (d) Rule 38

6. (a) Interpret any TWO of the following :
   (i) Restricted visibility
   (ii) Constrained by draft
   (iii) RAM vessel.

   Or

   (b) Draw and name the day symbol for (i) RAM vessel (ii) CBD (iii) NUC.

7. (a) Explain in your own words :
   (i) Safe speed
   (ii) Look out.

   Or

   (b) Write down rule number for
   (i) Overtaking vessel
   (ii) Head on situation
   (iii) Responsibility between vessel.

8. (a) Explain (i) Fishing vessel (ii) Seaplane.

   Or

   (b) Define (i) Power driven vessel (ii) Vessel not under command.

9. (a) Write the full form of :
   (i) AIS
   (ii) ATT
   (iii) GPS
   (iv) ECDIS
   (v) IALA
   (vi) VDR.

   Or

   (b) What is the difference between navigating in a narrow channel and in a TSS?

10. (a) What is a marine sextant?

Or

(b) What is the compass error and when do you apply it?

Part B $(5 \times 5 = 25)$

Answer **all** questions.

11. (a) What is known as "Crossing Situation" with respect to COLREG? In this case, who is known as "Stand On Vessel" and who is "Give Way Vessel"? What are their duties?

Or

(b) What are the responsibilities between vessels as per COLREG?

12. (a) Write a short note on :

(i) Dead reckoning position

(ii) Estimated position.

Or

(b) Write a short note on :

(i) Observed position

(ii) Deviation card.

13. (a) Name at least ten information that are included on charts.

Or

(b) List down four items which you obtain from Admiralty Catalogue.

14. (a) Explain compass error.

Or

(b) Explain color, range and characteristic of navigational light of power driven vessel of 100 meters above.

15. (a) Write short note on tide and how would it affect you during sailing.

Or

(b) What all information you would gather for your passage plan?

Part C $(3 \times 10 = 30)$

Answer **all** questions.

16. (a) What factors must be considered while selecting the basic equipment for chart work?

Or

(b) Discuss the factors which must be considered in placing the degree of reliability on navigational chart.

17. (a) (i) Name the categories of navigational charts.

(ii) Explain coastal and ocean going charts.

Or

(b) (i) What are special purpose chart?

(ii) Explain routing charts.

18. (a) Differentiate Mercator and Gnomonic chart.

Or

(b) Write the significance of "Date of Printing" and "Date of New Edition" on a navigational chart.

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B.Sc. DEGREE EXAMINATION, NOVEMBER 2017

Nautical Science

NAVAL ARCHITECTURE – II

(2016 onwards)

Time : 3 Hours Maximum : 75 Marks

Use hydrostatic particulars for solving M.V. Hindship problems.

Part A (10 × 2 = 20)

Answer **all** questions.

1. What is MCTC and state its formula?

2. What is LCB?

3. What is TPC and state its formula?

4. Define longitudinal metacentre.

5. Define LCF.

6. Do transverse bulkheads reduce free surface effects?

7. What are sounding pads and why are they fitted in tanks?

8. In which framing system used for bottom shell framing for a larger vessel of 120 m or more in length?

9. What is the prime function of Bilge keel?

10. How galvanic corrosion does takes place?

Part B $(5 \times 5 = 25)$

Answer **all** questions.

11. (a) A ship 150 metres long has a displacement of 7200 tonnes, and is floating upright on an even keel. When a weight of 60 tonnes, already on board, is shifted 24 metres forward, the trim is changed by 0.15 metres. Find the longitudinal metacentric height.

Or

(b) State different types of bulkheads.

12. (a) When a weight is shifted aft in a ship 120 metres long, it causes the ship's centre of gravity to move 0.2 metres horizontally and the trim to change by 0.15 metres. Find the longitudinal metacentric height.

Or

(b) Define propeller and principle behind the propeller.

13. (a) A ship of 6000 tonnes displacement has KB 3m, KM 6m and KG 5.5 m. Find the moment of statical stability at 10 degrees heel.

Or

(b) State different types of ventilators.

14. (a) Explain how anchor cable is connected to the bitter end with a neat sketch.

Or

(b) Three consecutive ordinates in a ship's water-plane, spaced 6 metres apart, are 14, 15 and 15.5 m, respectively. Find the area between the last two ordinates.

15. (a) State any 6 classification society.

Or

(b) Explain what are panting and pounding forces.

Part C $(3 \times 10 = 30)$

Answer **all** questions.

16. (a) A box shaped vessel 120 m × 20 m floats at an even keel draft of 4.5 m. 360 t of cargo is loaded 40 m abaft mid ship, 120 t of ballast is pumped out from 30 m abaft of midship. A weight of 60t is shifted from 20 m abaft midship to 40 m forward of mid ship. Calculate her final drafts, given KG = 5.1 m.

Or

(b) Draw a neat sketch of Midship section of double hull Bulk carrier and label the parts.

17. (a) M.V. Hind ship in condition No.2 has to load 340 t of cargo. Where with respect to the after perpendicular should this weight be loaded so that her after draft remains unaltered?

Or

(b) What is Statical stability Curve and enumerate the stability data you will get from this curve?

18. (a) A ship 150 metres long arrives at the mouth of a river with drafts 5.5 m F and 6.3 m A. MCT 1 cm 200 tonnes m. TPC 15 tonnes. Centre of flotation is 1.5 m aft of amidships. The ship has then to proceed up the river where the maximum draft permissible is 6.2 m. It is decided that SW ballast will be run into the forepeak tank to reduce the draft aft to 6.2 m. If the centre of gravity of the forepeak tank is 60 metres forward of the centre of flotation, find the minimum amount of water which must be run in and also find the final draft forward.

Or

(b) Explain about Hawse pipe and securing arrangement.